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Task Description:

Objectives: Abnormal immune reactions have occurred in some crew members during spaceflight, imposing significant threats on the success of the missions and health of the crew members. By subjecting mice to hindlimb unloading (HU), a ground-based model that simulates spaceflight, we have found that lymphocyte numbers are dramatically reduced and the immune response is significantly altered. Our previous results using this model and radiation, we proposed to further investigate the cellular and molecular mechanisms by which spaceflight affects the immune system using the HU model. Our original aims were to: 1) Elucidate the role of Fas and endogenous opioids in the modulation of the immune system by HU and radiation; 2) Determine the effect of HU and radiation on lymphocyte dynamics and immune responses, and 3) Determine whether HU and radiation promote the development of autoimmunity. All aims of this project remain same, except that we have adapted more advanced approaches in some experiments.
Progress: We have reported that HU causes a drastic apoptosis and depletion in various cell types in the spleen and thymus. Endogenous opioids and Fas-FasL interaction are important for HU-induced reductions in splenocytes, while corticosteroid mediates the reduction of lymphocytes in both the spleen and thymus. HU severely curtailed antigen-specific splenocyte proliferation, the delayed-type hypersensitivity response, and expansion of antigen-specific T cells. We analyzed splenocyte subpopulations from HU-treated mice to identify B cells, T cells and NK cells, and CD4, CD8 and CD25 T cell subpopulations. B cells were most significantly affected, although all major populations were reduced. In the thymus, CD4+CD8+ cells were particularly vulnerable. Surprisingly, CD4+CD25+ T cells were resistant to HU. Depletion of Tregs by anti-CD25 antibody prevented HU-induced lymphocyte reduction. Elimination of Treg in vitro prevented the inhibitory effect of HU on splenocyte proliferation and cytokine production. Thus, CD4+CD25+ Tregs play a critical role in lymphocyte homeostasis.

In collaboration with Dr. Denhardt and Kathryn Wang at Rutgers University, we examined the role of osteopontin (OPN), a bone matrix protein, in HU-induced lymphocyte reduction. Recent studies, however, have shown that this protein also plays a role in the immune system. We found that OPN-knockout mice are resistant to HU-induced reduction of splenocytes and thymocytes.

Since we found that free radicals are important in the apoptosis process in vitro, we tested the effect of free radical scavengers. Vitamin C provided in drinking water at 2 mg/ml for 10 days, protected the thymus from HU-induced involution. We also tested Dr. Ann Kennedy's antioxidant-rich food supplement on HU- and radiation-induced lymphocyte loss. Mice fed this diet for 2 weeks were protected from lymphocyte deletion induced by radiation. Another promising countermeasure is seaweed polysaccharide extract. We tested a polysaccharide extract from the brown seaweed, Laminaria japonica (part of the Eastern diet for centuries), using a commercial product from NaturoDoc. When fed at 100 mg/mouse/day for 3 wk, this seaweed extract inhibited radiation-induced lymphocyte loss in the spleen. In collaboration with Dr. Marcelo Vazquez of Brookhaven National Laboratory, we investigated the long term effect of particle radiation on immune cells. At 6 month after various doses of particle radiation, few consistent cahnges were apparent in the populations of immune cells examined, except for a reduction in B cell numbers.

The degree of stress in an individual is difficult to quantify due to the lack of a specific marker. Using proteomics, we found that transferrin and haptoglobin were consistently increased in the serum of mice after HU. To extend this finding to the astronaut population, we are collaborating with Drs. Duane Pierson and Satish Mehta of Johnson Space Center, who have provided serum samples from astronauts pre-and post-flight. We found increased haptoglobin in astronaut serum in comparison to a control group. Thus, serum levels of haptoglobin may provide a biomarker to quantitatively measure stress levels in astronauts.

Future Plans: Based on our new data, we will focus our research in the following areas in the coming year.

1. We will continue our effort to find reagents, especially nutritional food supplements or common pharmaceutical approaches, to prevent lymphocyte apoptosis and help to maintain immune homeostasis. We will focus on vitamin C as well as its combination with other supplements such as melatonin and kelp extract. We will also examine the effect of neutralizing OPN. We will continue to collaborate with Dr. Ann Kennedy in this area. 2. Develop immune parameters that will predict the status of the immune system. We will examine the reactivity of lymphocyte at different times after activation and rescue by different reagents. We will employ the Luminex multi-plexed assay system to detect various cytokines. 3. We will determine mechanisms of synergy between HU and radiation in lymphocyte reduction. We will first focus on the role of free radicals. We will continue our collaboration with Dr. Marcelo Vazquez to examine the short-term effects of particle radiation and in combination with HU. 4. We will further validate the changes of haptoglobin in astronauts by examining more serum samples. 5. We will test if HU and radiation affect the development of autoimmunity. We will examine changes in thymic negative selection of autoreactive T cells, and alterations in apoptotic cell clearance in the periphery. We have obtained some preliminary data showing that apoptotic cells can induce immune tolerance through regulation of dendritic cells, while necrotic cells promote immune responses. Since a large number of cells undergo apoptosis during HU, some of these cells may persist and become necrotic.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

We hope to gain an understanding of the mechanisms by which stress affects immune homeostasis. With this information, it may be possible to devise dietary and lifestyle changes which reduce the deleterious effects of stress on the immune system. Such changes may lead to the improvement of public health. We also hope to identify food supplements, such as vitamin C and seaweed extracts that reduce the health impact of stress, bringing benefit to the general population, especially those who suffer from high levels of occupational or life stress.
By determining the role of cellular apoptosis in mediating the deleterious effects of stress on the immune system we may be able to aid in the development of protocols for the maintenance of immune tolerance for clinical use in organ transplant or autoimmune disease.

We are studying the role of a specific subset of T lymphocytes called CD4+CD25+ regulatory T cells which seem to be involved in the deleterious effect of stress on the immune system. Learning how these cell exert their effects may lead to potential interventions to prevent the negative health impacts of autoimmunity, cancer, and other diseases that arise from an improperly regulated immune system. It is difficult to measure the level of stress that an individual experiences for the purpose of allowing appropriate intervention. Identification of particular serum proteins whose concentration is related to the stress level could serve as a biomarker for the assessment of stress levels for medical or occupational purposes.

Task Progress:	1) Elucidate the role of Fas and endogenous opioids in the modulation of the immune system by HU and radiation. We have found that the synergestic effect of radiation and HU is also exerted through Fas since anti-FasL could block this effect. 2) Determine the effect of HU and radiation on lymphocyte dynamics and immune responses. Depletion of Tregs by anti-CD25 antibody prevented HU-induced lymphocyte reduction. Osteopontin (OPN) deficient mice are resistant HU-induced lymphocyte reduction, suggesting that elimination of OPN may provide a countermeasure. Vitamin C and antioxidant-rich food supplement can inhibit HU- and radiation-induced lymphocyte loss. Seaweed polysaccharide extract can inhibit radiation-induced lymphocyte loss. In collaboration with Dr. Marcelo Vazquez of Brookhaven National Laboratory, we investigated the long term effect of particle radiation on immune cells. At 6 month after various doses of particle radiation, we did not find significant change in lymphocytes. This might be too long after radiaiton and the dose is too low. New experiments are planned. 3) Determine whether HU and radiation promote the development of autoimmunity. We have examined the effect of apoptotic and necrotic cells on immune response. We found that unlike apoptotic cells, necrotic cells are stimulators of immune responses. 4) Studies not in the original aim. The degree of stress in an individual is difficult to quantify due to the lack of a specific marker. Using proteomics, we found that transferrin and haptoglobin were consistently increased in the serum of mice after HU. To extend this finding to the astronaut population, we are collaborating with Drs. Duane Pierson and Satish Mehta of Johnson Space Center, who have provided serum samples from astronauts pre-and post-flight. We found increased haptoglobin in astronaut serum in comparison to a control group. Thus, serum levels of haptoglobin may provide a biomarker to quantitatively measure stress levels in astronauts.
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